

Electrochemistry

Chemical Energy $\rightleftharpoons$ Electrical Energy

Electrochemical cell

Galvanic cell
or
voltaic cell

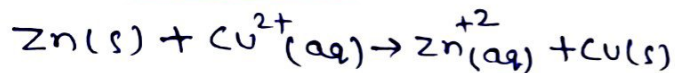
- Chemical $\rightarrow$ Electrical
- Spontaneous redox Reaction ($\Delta G < 0$)

Electrolytic cell

- Electrical $\rightarrow$ chemical
- Non-Spontaneous redox reaction ($\Delta G > 0$)

Working of Galvanic cell Eg $\rightarrow$ Daniel cell

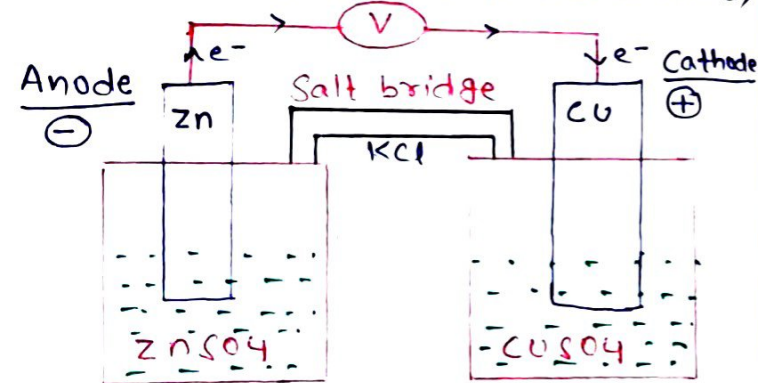
- At Anode (Oxidation) $\rightarrow \text{Zn} / \text{Zn}^{2+}$
- At Cathode (Reduction) $\rightarrow \text{Cu}^{2+} / \text{Cu}$
- Cell reaction:-



- Current flow:-

Anode $\rightarrow$ Cathode (internal circuit)

Cathode $\rightarrow$ Anode (external circuit)



L $\rightarrow$ Left

R $\rightarrow$ Right

O $\rightarrow$ Oxidation

R $\rightarrow$ Reduction

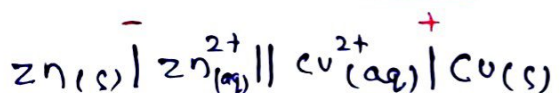
A $\rightarrow$ Anode

C $\rightarrow$ Cathode

N $\rightarrow$ Negative

P $\rightarrow$ Positive

★ Cell Representation:-



$\hookrightarrow$ salt bridge

Salt bridge & its function

- It is U-shaped glass tube which contains jelly like substance, agar-agar with an electrolyte like KCl, KNO₃, NH₄NO₃ etc.
- Mobility (speed) of cation $\approx$ mobility of Anion.
- If Ag⁺, Pb²⁺, Tl⁺ present in solⁿ the KCl can not use in salt bridge.

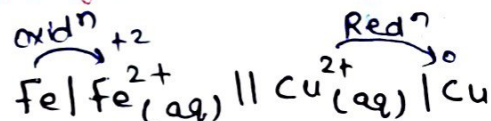
★ Function:-

- It connects the solution of two half cell to complete the circuit.
- It maintains the electrical neutrality of the solution.
- To complete inner circuit without mixing of two solution
- To reduce liq-liq junction potential.

★ Cell potential difference (E°_{cell})

$$E^\circ_{\text{cell}} = (E^\circ_R)_C - (E^\circ_R)_A$$

Q $\rightarrow$ For galvanic cell find E°_{cell} ?



Given

$$E^\circ_{\text{Fe} | \text{Fe}^{2+}} = +0.44 \text{ V}$$

$$\hookrightarrow E^\circ_{\text{Fe}^{2+} | \text{Fe}} = -0.44 \text{ V}$$

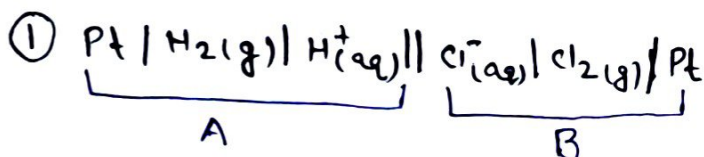
$$E^\circ_{\text{Cu}^{2+} | \text{Cu}} = +0.34 \text{ V}$$

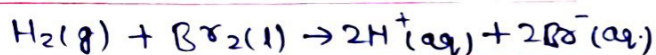
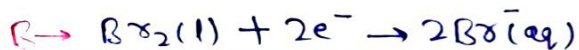
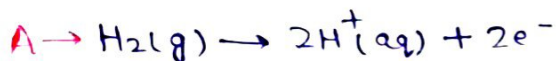
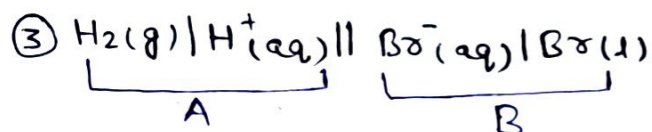
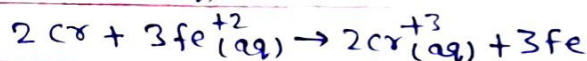
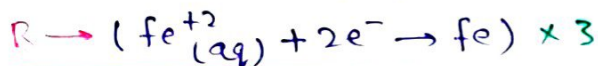
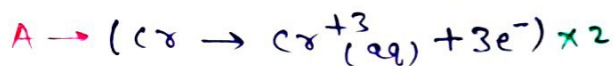
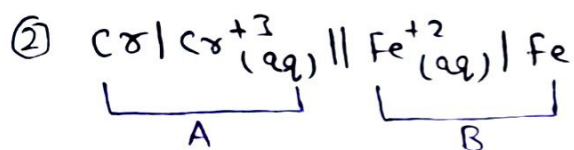
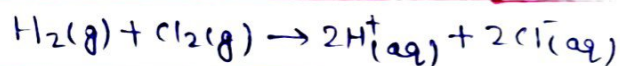
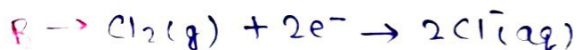
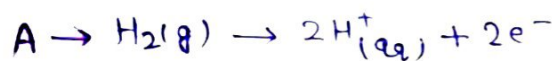
$$E^\circ_{\text{cell}} = (E^\circ_R)_C - (E^\circ_R)_A$$

$$= (+0.34) - (-0.44)$$

$$= 0.78 \text{ V.}$$

Q $\rightarrow$ Give the anode reaction, cathode rxn & complete cell reaction for the following?

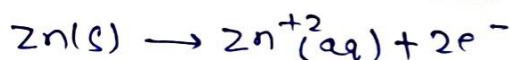




Electrode potential

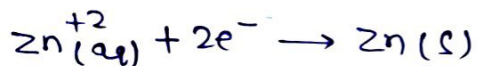
↳ Potential difference b/w electrode (metal rod) & electrolytic solⁿ.

① Oxidation Potential



$$E_{oxid} [E_{M/M^+} = O.P.]$$

② Reduction Potential



$$E_{red} [E_{M^+/M} = R.P.]$$

- Acc. to IUPAC potential of metal is equal to reduction potential of metal.

★ Electrode potential is depend on

- Nature of Electrode
- Temp. pressure condition
- Concⁿ of the solⁿ.

Standard condition

- standard condition

$$P_{(gas)} = 1 \text{ atm}$$

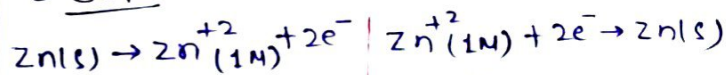
$$T = 25^\circ\text{C} \text{ (at any Temp)}$$

$$\text{Conc}^n \text{ of all ions} = 1M$$

$$E.P. = SEP$$

SEP

① SOP

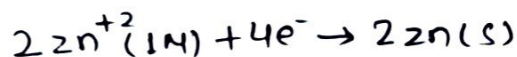


$$E_{Zn/Zn^{+2}} = 0.76V$$

$$E_{Zn^{+2}/Zn} = -0.76V$$

$$SRP = -SOP$$

- ★ SEP does not depend on stoichiometry coefficient of reⁿ.



$$L \cdot [E^{\circ}_{red}] = -0.76V$$

- ★ Value of SEP of hydrogen electrode is assuming to be zero & it is also called Reference electrode.

Representation - $Pt, H_2(g) | H^+_{(1M)}$

↳ for SHE, electrode potential is assumed to be zero at all Temp.

$$SOP = -SRP = 0 \text{ for SHE (standard hydrogen electrode)}$$

★ Application of electrochemical series [ECS]:-

→ Reactivity of Metals

- In ECS, reactivity of Metal ↓ as from top to bottom.
- Li is most reactive metal

$$L \cdot \text{Reactivity of Metal} \propto \frac{1}{SRP} \propto SOP$$

→ Reactivity of Non-metals

- In ECS, reactivity of non-metals increases from top to bottom.
- F_2 is most reactive non-metal.
- Reactivity of non-metal $\propto$ SRP $\propto \frac{1}{SOP}$



Reducing nature ↓ se from T → B

$\left[\begin{array}{l} \text{Reducing nature} \\ \text{Reducing power} \\ \text{Reducing Agent} \end{array} \right] \propto \frac{1}{SRP} \propto SOP$

Oxidising nature ↑ se from T → B

$\left[\begin{array}{l} \text{Oxidising nature} \\ \text{Oxidising power} \\ \text{Oxidising Agent} \end{array} \right] \propto SRP \propto \frac{1}{SOP}$

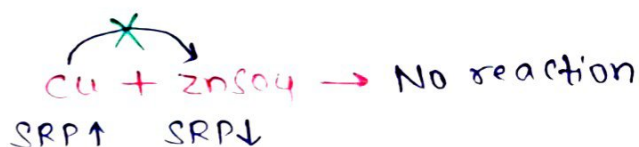
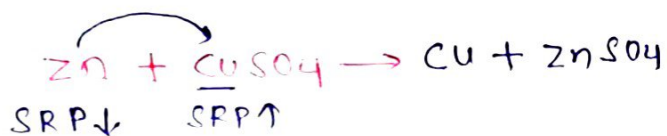
Electrochemical Series

Le	Li		Ni	neeche kahe
Kar	K		Sn	Suno
Barah	Ba	$\uparrow E^\circ_R = -ve$	Pb	pyar
Sardar	Sr	$E^\circ_R = 0 \rightarrow$	H	hai
ka	Ca	$E^\circ_R = +ve$	Cu	kyun
naam	Na	$\downarrow$	I ₂	aai
Maange	Mg		Ag	aaj
Ali	Al		Hg	Hongkong me
Mahan	Mn		Br ₂	Britney
Zaan	Zn		Pt	Patai
Kar	Cr		O ₂	aur
fenke	Fe		Cl ₂	Kilobhar
C.D	Cd		Au	Sona laai
Koi	Co		F ₂	Free mein

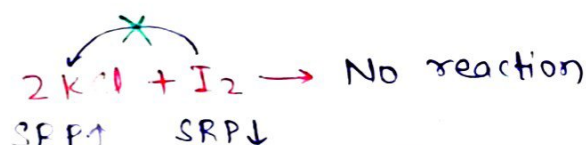
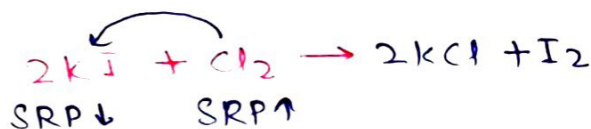
* Displacement rxn in solution

→ More reactive metal/non-metal displaces less reactive metal/non-metal in their solution.

Metal to Metal displacement rxn



Non-metal to non-metal displacement reaction →



Nernst Equation

→ It is a relation b/w electrode potential, emf of cell, temp. and concⁿ of ions in solⁿ represented by the equation.

$$E_{cell} = E^\circ_{cell} - \frac{RT}{nF} \ln \frac{[P]}{[R]}$$

At 25°C

$$E_{cell} = E^\circ_{cell} - \frac{0.059}{n} \log_{10} \frac{[P]}{[R]}$$

E_{cell} = EMF of cell at new concⁿ.

E°_{cell} = standard emf of cell
(1M, 25°C & 1atm).

R = Gas constant = 8.314 J K⁻¹ mol⁻¹.

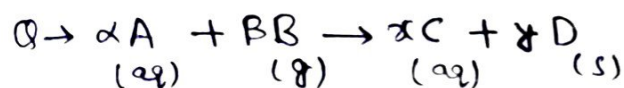
T = Absolute temp.

n = no. of electrons loose or gain.

F = Faraday constant = 96500 C.

$[P]$ = concⁿ of product in mol L⁻¹.

$[R]$ = concⁿ of reactants in mol L⁻¹.



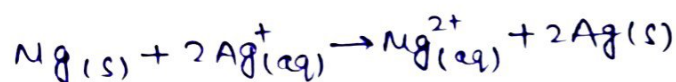
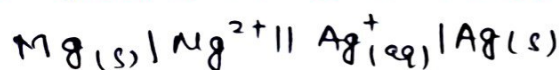
$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.06}{n} \log \frac{[C]^x}{[A]^a [B]^b}$$

At equilibrium

$$E_{\text{cell}} = 0, Q = K_{\text{eq}}$$

$$[E^{\circ}_{\text{cell}} = -\frac{0.06}{n} \log K_{\text{eq}}]$$

Q → Find EMF of cell, $E^{\circ}_{\text{cell}} = 3.17V$



$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.06}{n} \log \frac{[Mg^{2+}]}{[Ag^+]^2}$$

$$= 3.17 - \frac{0.06}{n} \log \left[\frac{10^{-1}}{(10^{-3})^2} \right]$$

$$= 3.02V$$

★ For the cell -



at 298K

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log \frac{[M]^m [N]^n}{[A]^a [B]^b}$$

$$= E^{\circ}_{\text{cell}} - \frac{0.06}{n} \log \frac{[\text{Oxi}]}{[\text{Red}]}$$

$$E_{\text{cell}} = +ve \text{ if } C_2 > C_1$$

Gibbs Energy

$$\Delta G = -W_{\text{electrical}}$$

$$\Delta G = -nFE_{\text{cell}}$$

n → no. of transfer e⁻

f → 96500 C

Case (I) If $E_{\text{cell}} = 0$

then $\Delta G = 0$ at equilibrium state
($R_f - R_b$)

Case (II) If $E_{\text{cell}} > 0$, then

$\Delta G < 0$, spontaneous.

Case (III) If $E_{\text{cell}} < 0$, then

$\Delta G > 0$, non-spontaneous.

$$E^{\circ}_{\text{cell}} = \frac{0.06}{n} \log_{10} K_{\text{eq}}$$

[K_{eq} = equilibrium constant]

$$\Delta G^{\circ} = -2.303RT \log_{10} K$$

Quantitative aspects of electrolysis

① First law of Faraday →

- The amount of Metal deposit at respective electrode is proportional to total charge flow through solⁿ of electrolyte.

$$w \propto Q$$

$$\therefore Q = It$$

$$w = zQ$$

$$z = \frac{E}{F} = \frac{E}{96500}$$

$$w = zIt$$

z = electrochemical equivalent.

$$w = \frac{EIt}{F}$$

$$w = \frac{MIt}{n \cdot f \times F}$$

$$\left[E = \frac{At \cdot \text{mass}}{n \cdot f} \right]$$

$$\frac{w}{M} = \text{mole} = \frac{It}{n \cdot f \times F}$$

$$n \times n \cdot f \times F = It$$

② 2nd Law of Faraday →

Q = same

$$w = \frac{EIt}{F}$$

$$\frac{w}{E} = \text{constant}$$

$$\frac{w_1}{w_2} = \frac{z_1}{z_2} = \frac{e_1}{e_2}$$

Electrolytic conductance

[1] Resistance (R) $\rightarrow$

Ohm's law, $V \propto I$

$$V = IR$$

$$R = \frac{V}{I} = \frac{\text{Volt}}{\text{Amp}} = \Omega$$

[2] Resistivity (ρ)

$$R \propto \frac{l}{A}, \quad R = \rho \frac{l}{A}$$

$$\beta = \frac{RA}{l} \quad (0.44 - 0.4)$$

[3] Conductance (G)

$$\boxed{G = \frac{1}{R}} \quad \begin{array}{l} (ohm)^{-1} \text{ or } who \\ (\Omega)^{-1} \text{ or } \omega \\ \text{or Siemens (S)} \end{array}$$

[4] Conductivity (κ)

$$K = \frac{1}{f}, \quad f = \frac{RA}{l}$$

$$K = \frac{1}{R} \times \frac{l}{A}$$

for one cell, $\frac{1}{A} = \text{cell const. (G}^*)$

$$K = \frac{1}{R} \times 5^* \Rightarrow K = 5 \times 5^*$$

unit of k is $\text{ohm}^{-1} \text{cm}^{-1}$

(5) Molar conductivity (λ_m) or
Molar conductance $\rightarrow$

$$\lambda_m = k \times v_m \text{ (m/s)}$$

$$\lambda_m = K \times \frac{1000}{M} \rightarrow \text{Molarity}$$

$$\text{Unit} \rightarrow \lambda u. K \times V u$$

$$\lambda_{\text{max}} = 0.69 \times 10^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

(b) Equivalent conductivity $[\lambda_{eq}]$

$$\lambda_{eq} = k \times V_N(w)$$

$$\lambda_{eq.} = k \times \frac{1000}{N} \rightarrow \text{Normality}$$

Unit $\rightarrow \lambda e_q = 0 \hbar \omega^{-1} c \omega^2 e_q^{-1}$

★ Relation b/w λ_m & λ_{eq}

$$\lambda_m = \lambda_{eq} \times n - f$$

Factors affecting electrolytic Conductance :-

(1) Nature of electrolyte

- For strong electrolyte
 $\alpha = 1$ [conductance $\uparrow$]
- ~~Inter ionic attraction force (IIAF)~~
- For weak electrolyte
 $\alpha = 1$ [conductance $\downarrow$]

(2) Inter ionic attraction force (IIAF)

- IIAf ($\downarrow$), Mobility ($\uparrow$), Conductance ($\uparrow$)

(3) Effect of temperature

- Temp($\uparrow$), IIAF($\downarrow$)
- Mobility($\uparrow$), Conductance($\uparrow$)

(4) solvation or hydration of ion

- Association of ion with solvent molecule is called solvation.
- Solvation ($\uparrow$), Mobility ($\downarrow$), Conductance ($\downarrow$)

Mobility $\rightarrow H^+ > OH^- > \oplus > \ominus$

(5) Effect of dilution

(i) $G \uparrow$, $V \uparrow$

S.E	W.E
$\alpha = 1$	$\alpha < 1$
$V \uparrow$	$V \uparrow [\alpha \propto \sqrt{V}]$
Mobility $\uparrow$	no. of ions $\uparrow$
$G \uparrow$	$G \uparrow$

(ii) $K \downarrow$, $V \uparrow$

- Specific conductance of electrolytic solⁿ is decrease with increase the dilution because no. of ion present in 1 cm^3 solⁿ is decreases.

(iii) Molar conductance & equivalent conductance of electrolytic solⁿ is increase with increase the dilution because mobility of ion (strong electrolyte) or no. of ion (weak electrolyte) is increase.

Debye Huckel onsager equation

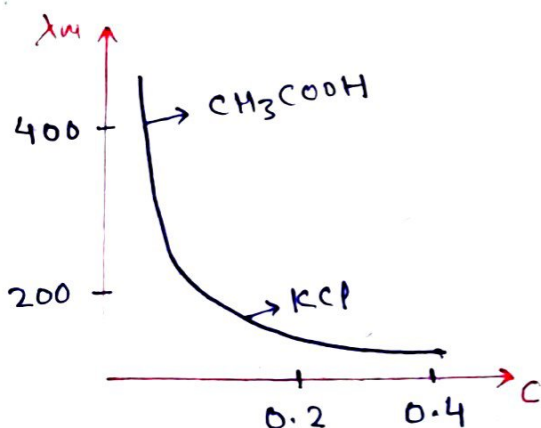
$$\Lambda_m = \Lambda_m^\infty - b\sqrt{C} \quad \text{only for strong electrolyte.}$$

Where,

Λ_m = molar conductance at concⁿ C

Λ_m^∞ = molar conductance at ∞ dilution.

b = constant & its value is same for a particular type of electrolyte.



Kohlrausch Law

↳ Independent of migration of ions.

- The equivalent conductance of electrolyte at ' ∞ ' dilution is equal to sum of equivalent of cation & Anion at ' ∞ ' dilution.

- All electrolyte [weak or strong]

$$\lambda_{eq}^\infty = \lambda_{eq}^\infty(c) + \lambda_{eq}^\infty(a)$$

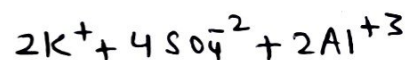
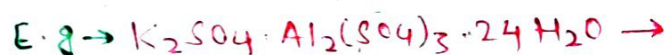
$$\lambda_M^\infty = x \lambda_M^\infty(c) + y \lambda_M^\infty(a)$$

$$\lambda_{eq}^\infty = \frac{1}{x} \lambda_M^\infty(c) + \frac{1}{y} \lambda_M^\infty(a)$$

$$\lambda_M^\infty = \lambda_{eq}^\infty \times n-f$$

$x \rightarrow$ Mole of cation, $C \rightarrow$ cation

$y \rightarrow$ Mole of anion, $a \rightarrow$ Anion



$$\lambda_M^\infty = 2\lambda_M^\infty(K^+) + 4\lambda_M^\infty(SO_4^{2-}) + 2\lambda_M^\infty(Al^{3+})$$

$$n-f = 8 \quad \lambda_M^\infty = \lambda_{eq}^\infty \times n-f$$

$$\lambda_{eq}^\infty = \frac{\lambda_M^\infty}{n-f} = \frac{\lambda_M^\infty}{8}$$

Application of Kohlrausch law

① Determination of Molar conductance of weak electrolyte at ' ∞ ' dilution

$$\lambda_M^\infty(CH_3COOH) = \lambda_M^\infty(CH_3COONa) + \lambda_M^\infty(HCl) - \lambda_M^\infty(NaCl)$$

$$\lambda_M^\infty(BaCl_2) = \lambda_M^\infty(Ba(OH)_2) - 2\lambda_M^\infty(NaOH) - 2\lambda_M^\infty(NaCl)$$

$$\Rightarrow \lambda_M^\infty(Na_2C_2O_4) = x_1$$

$$\lambda_M^\infty(H_2SO_4) = x_2$$

$$\lambda_M^\infty(Na_2SO_4) = x_3$$

$$\lambda_M^\infty(H_2C_2O_4) = x_1 + x_2 - x_3$$

$$\Rightarrow \lambda_M^\infty(\text{H}_2\text{C}_2\text{O}_4) = \lambda_M^\infty(\text{Na}_2\text{C}_2\text{O}_4) + \lambda_M^\infty(\text{H}_2\text{SO}_4) - \lambda_M^\infty(\text{Na}_2\text{SO}_4)$$

② Determination of degree of dissociation of electrolytes $\rightarrow$

$$\alpha = \frac{\lambda_M^c}{\lambda_M^\infty} = \frac{\lambda_M \text{ at given concn}}{\lambda_M \text{ at } \infty \text{ dilution}} \text{ or } \frac{\lambda_{eq}}{\lambda_{eq}^\infty}$$

$$\alpha = \frac{k \times \frac{1000}{M}}{\lambda_M^\infty}$$

③ Determination of dissociation constant of weak electrolyte $\rightarrow$



$$t=0 \quad C \quad 0 \quad 0$$

$$t_{eq} \quad C - C\alpha \quad C\alpha \quad C\alpha$$

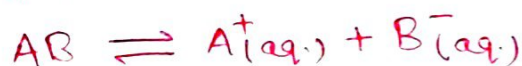
$$C[1-\alpha]$$

$$K_a = \frac{C\alpha^2}{(1-\alpha)}$$

If $\alpha \leq 5\%$, then

$$K_a = C\alpha^2$$

④ Determination of solubility of saturated soln of electrolytes $\rightarrow$



$$K_{sp} = S^2$$

$$\lambda_M = k \times \frac{1000}{M}$$

• Saturated soln = ∞ dilution

$$S = M$$

$$\lambda_M = \lambda_M^\infty$$

$$\lambda_M^\infty = k \times \frac{1000}{S}$$

$$S = k \times \frac{1000}{\lambda_M^\infty}$$

Corrosion $\rightarrow$ Process of Metal surface into salt like oxides, sulphides, due to atmospheric gases.

* Rust $\rightarrow$ It is hydrated ferric oxide ($\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$)

* Factors affecting which enhance corrosion

- ① Presence of impurities in metal.
- ② Presence of Moisture.
- ③ Presence of electrolyte.

$\rightarrow$ Corrosion can be prevented by Barrier protection, Sacrificial protection.

Battery & cell

$\Rightarrow$ Batteries $\rightarrow$ Source of electrical energy which may have one or more cells connected in series.

① Primary Batteries $\rightarrow$ Rxn occurs only once & after use over a period of time, battery becomes dead & can't be reused.

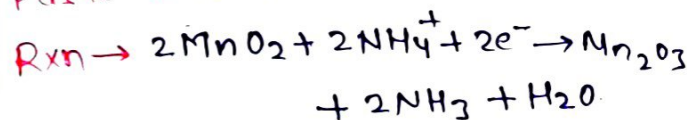
eg $\rightarrow$ Dry cell and Mercury cell.
[Leclanche cell]

① Dry cell

Anode $\rightarrow$ Zinc container

Cathode $\rightarrow$ Graphite rod

Paste $\rightarrow \text{NH}_4\text{Cl} + \text{ZnCl}_2$



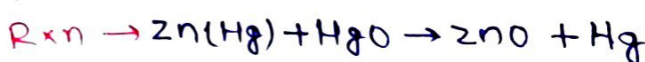
cell potential $\rightarrow 1.25\text{V} - 1.5\text{V}$

② Mercury cell

Anode $\rightarrow \text{Zn-Hg amalgam}$

Cathode $\rightarrow \text{HgO} + \text{C}$

Paste $\rightarrow \text{KOH} \& \text{ZnO}$



cell potential $\rightarrow 1.35\text{V}$

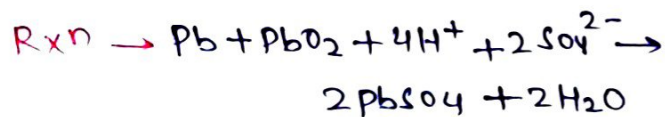
② Secondary Batteries:- cell can be recharged & reused.

eg → Lead storage battery & Nickel cadmium cell

① Lead storage

Anode → Spongy lead

Cathode → Pb with PbO_2

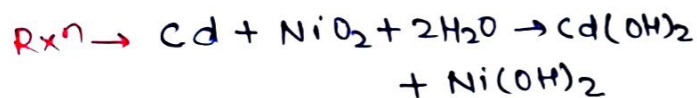


- When recharged, cell rxn are reversed.

② Nickel - Cadmium storage cell

Anode → Cadmium

Cathode → Metal + NiO_2



cell potential → 1.4 V

- Nickel-cd cell has longer life than the lead storage cell.

Fuel cells →

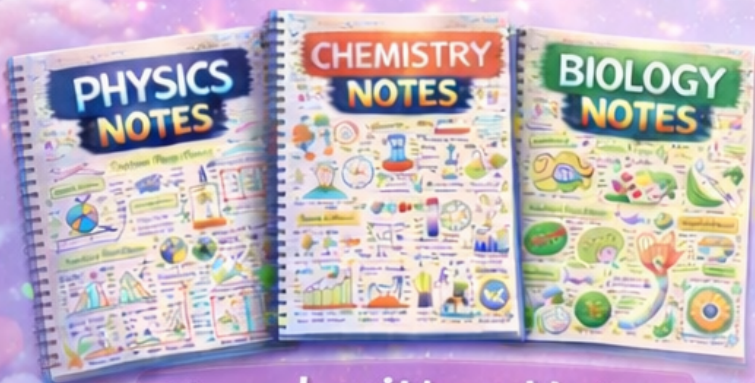
↳ Galvanic cells which are energy of combustion of fuel like H_2 , CH_4

↳ pollution free & have high efficiency.

eg → Hydrogen, oxygen, fuel cells (H_2O_2 fuel cell) ..

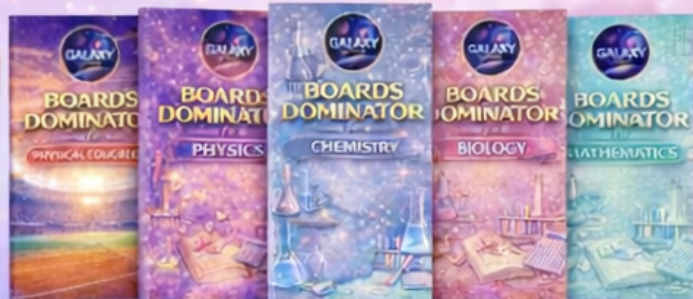
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